
LEARNING BEYOND GRADIENTS: RESOURCES, CERTIFICATES, AND LIMITS OF LLM-GUIDED ARTIFACT SEARCH

Anonymous authors

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ABSTRACT

A rapidly growing family of systems improves at tasks without touching model weights: an outer loop proposes edits to a textual or programmatic *artifact*—a prompt, a skill document, an agent harness, a training script—scores each candidate with a noisy evaluator, and keeps what works, with a frozen large language model (LLM) acting as the proposal distribution. Instances include reflective prompt evolution, self-evolving skills and contexts, harness optimization, automated agent design, and evolutionary code search. We give a single formal model that contains these systems and prove that its behavior is governed by a small number of measurable resources: the *prior mass* the proposer places on good artifacts, which controls hitting time Levin-style and whose conditioning boost γ we show is estimable from samples at finite cost; the *information content of feedback*, where scalar rewards can be exponentially slower than structured traces; the *noise-gap ratio* of the evaluator, where a single-evaluation greedy ratchet fed pure noise reports a realized median improvement of 3.4σ after 2,000 null edits (climbing toward the $\sigma\sqrt{2\ln t}$ ceiling of 3.9σ) and accepts ≈ 8.2 of them, while $O((\sigma^2/\Delta^2)\log(B/\delta))$ repeated evaluations per decision certify a whole run; the *deceptiveness* of the landscape, where archive-based selection is exponentially faster than tolerant greedy search; and the *description length* of the artifact, through an Occam bound whose U-shaped risk predicts both context collapse and validation overfitting. Beyond analyzing each resource in isolation, we prove a *composed certified-improvement theorem* that requires the assembled object: a gated loop run under all three statistical resources at once finds and soundly certifies a near-optimal artifact in $O((\sigma^2/\Delta^2)\ln(B/\delta)\ln(1/\delta)/(\gamma q_0(G_\varepsilon)))$ evaluator calls, with false rejection of true improvements appearing as an explicit slowdown factor. We map each result to documented behaviors and failures of deployed systems, grounding the selection results in the archival evaluation-statistics literature, and we state falsifiable scaling predictions together with a pre-registered protocol for measuring γ on a real proposer.

1 INTRODUCTION

The dominant formal account of machine learning is gradient-based: define a differentiable loss, follow its gradient in weight space. Yet many striking recent gains in LLM-based systems involve no weight updates at all. GEPA evolves prompts by reflecting on execution traces and reports outperforming the RL algorithm GRPO with up to $35\times$ fewer rollouts (Agrawal et al., 2026). Meta-Harness rewrites the scaffolding code around a frozen model and beats hand-built context-management systems (Lee et al., 2026). SkillOpt edits a skill document under held-out acceptance tests and lifts frozen-agent accuracy by double digits (Yang et al., 2026). AlphaEvolve and FunSearch evolve programs against machine-checkable evaluators and produce new mathematics (Novikov et al., 2025; Romera-Paredes et al., 2024). Karpathy’s *autoresearch* loop edits a real training codebase overnight (Karpathy, 2026); Heuristic Learning systems patch executable heuristics from replayed failures (Weng, 2026); the Darwin Gödel Machine and ADAS search over agent designs themselves (Zhang et al., 2025a; Hu et al., 2025).

Each of these systems was presented with its own vocabulary—reflection, evolution, context engineering, self-improvement—and each paper’s analysis, where present, is empirical. This leaves basic questions open. Why are these loops so sample-efficient when classical program search is hopeless? When does a scalar reward suffice and when are traces necessary? Why do noisy single evaluations produce celebrated improvements that later fail to replicate—a failure mode documented for a decade in the evaluation-statistics literature (Henderson et al., 2018; Dodge et al., 2019; Agarwal et al., 2021) and recently relived by autonomous research loops? Why do greedy prompt optimizers overfit their validation sets while bounded-edit optimizers generalize? Why do archives and Pareto frontiers keep appearing in independently developed systems?

This paper. We answer these questions in a single model. An *artifact-search system* (Definition 2.1) is a tuple $(\mathcal{A}, F, q, \text{sel}, \mathcal{M})$: a countable space of textual/programmatic artifacts, a noisy evaluator, an LLM proposal kernel conditioned on execution feedback and memory, a selection rule, and an accumulating history. Learning updates the artifact; the model weights never move. Within this model we prove:

(1) Prior mass governs hitting time, and the boost is measurable (§3). The expected number of proposals before an ε -good artifact appears is the inverse of the mass the proposer assigns to the good set; adaptive trace-conditioning enters as a multiplicative boost γ (Theorem 3.2), with a matching lower bound (Proposition 3.3). New in this version: γ is not a free parameter—Proposition 3.5 shows the plug-in estimator from K verified proposals per condition certifies γ to relative error η once $K = O(\ln(1/\delta)/(\eta^2 q_{\min}^2))$, and Appendix I gives a concrete measurement protocol.

(2) Feedback is a channel whose information content separates learning rates (§4). Identification with binary success feedback requires $\Omega(2^L)$ queries on L -bit artifact classes, while feedback naming the first failure needs at most L (Theorem 4.2); a transcript-counting bound interpolates (Theorem 4.1). This is a self-contained worst-case companion to the eluder-dimension theory of learning from language feedback (Xu et al., 2025).

(3) Noise makes greedy ratchets unsound, quantifiably (§5). A single-evaluation ratchet fed pure-noise edits reports a diverging best score— $\mathbb{E}[m_t] \leq \sigma\sqrt{2\ln t}$, attained asymptotically for Gaussian noise—while accepting $H_t = \ln t + O(1)$ null edits (Theorem 5.1); $N = O((\sigma^2/\Delta^2)\log(B/\delta))$ repeated evaluations per decision certify a whole B -step run (Proposition 5.2, Corollary 5.3). The constants in this section were audited end-to-end against our committed code (§9).

(4) Deception separates greedy from archive selection (§5). On an explicit deceptive landscape with a *unique* optimum, strict greedy search never arrives, ε -tolerant greedy needs $\exp(\Omega(k))$ steps, and an archive rule arrives in $O(k^2)$ (Theorem 5.4)—the formal content of the Pareto frontiers, archives, and lineages recurring across GEPA, ADAS, and the DGM.

(5) The resources compose: certified improvement at the product rate (§6). The central new result of this version. Theorem 6.1 couples prior mass, evaluator noise, and gated selection in one statement: under a margin condition, a fully gated loop makes *no* false accept and certifies an ε -good artifact within $O(\ln(1/\delta)/(\gamma q_0(G_\varepsilon)))$ proposals and $O((\sigma^2/\Delta^2)\ln(B/\delta)\ln(1/\delta)/(\gamma q_0(G_\varepsilon)))$ evaluator calls, with the false-reject slowdown explicit. No proper subset of the model’s components supports the statement.

(6) The artifact is a compressed posterior; description length controls generalization (§7). An Occam/PAC-Bayes bound yields a U-shaped risk in artifact length that simultaneously explains *context collapse* (too short) and validation overfitting (too long) (Theorem 7.1, Corollary 7.2).

(7) Proxy gaps bound the whole enterprise (§8). Measured progress transfers to true progress up to twice the evaluator’s bias sup-norm; selection harvests bias at the optimizer’s curse rate $\sigma_b\sqrt{2\ln m}$; and if the loop can modify its own evaluator, an in-model construction drives measured score to the top of the range while true score traverses to the bottom—no nonvacuous transfer guarantee survives (Proposition 8.1).

Sections 9–10 provide simulations (each arm an executed algorithm) and situate the framework; proofs are in the appendices. Our aim follows Bowling et al. (2023): not to introduce a new algorithm, but to make precise what a fast-moving empirical literature is doing, so that its successes and failures become predictions rather than surprises—and, where the predictions are quantitative, to state them in falsifiable form (§11).

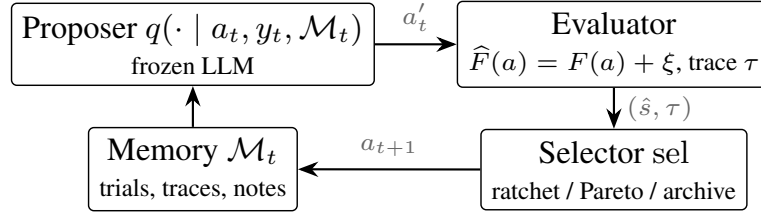


Figure 1: The artifact-search loop. Weights are frozen everywhere; learning is the trajectory $a_0, a_1, \dots$ through artifact space. Systems differ only in $\mathcal{A}$, the feedback alphabet, and sel (Table 1).

Table 1: Deployed systems as artifact-search systems. (sc. = scalar score; tr. = execution trace.) Sources marked $\dagger$ are informal (blog posts or code repositories) rather than archival; we use them as illustrations, never as the sole support of a claim.

System	Artifact $\mathcal{A}$	Feedback $\mathcal{Y}$	sel
GEPA (Agrawal et al., 2026)	prompts	sc. + tr.	instant
SkillOpt (Yang et al., 2026)	skill doc	sc. + tr.	gated
ACE (Zhang et al., 2025b)	context/playbook	tr.	curate
Meta-Harness (Lee et al., 2026)	harness code	sc. + raw history	ratche
autoresearch † (Karpathy, 2026)	training code	sc. (noisy)	greedy
Heuristic Learning † (Weng, 2026)	heuristic code	tr. (logs, video)	gated
ADAS / DGM (Hu et al., 2025; Zhang et al., 2025a)	agent programs	sc. + tr.	archiv
FunSearch / AlphaEvolve (Romera-Paredes et al., 2024; Novikov et al., 2025)	programs	sc. (verified)	island
OPRO / APE / PromptBreeder (Yang et al., 2024; Zhou et al., 2023; Fernando et al., 2023)	prompts	sc.	popul
Voyager (Wang et al., 2024)	skill library	tr.	additiv

2 ONE LOOP, MANY NAMES: THE FORMAL MODEL

Definition 2.1 (Artifact-search system). An *artifact-search system* is a tuple $\mathcal{S} = (\mathcal{A}, F, q, \text{sel}, \mathcal{M})$ where (i) $\mathcal{A}$ is a countable set of *artifacts* (finite strings: prompts, skill documents, programs, harness code); (ii) $F : \mathcal{A} \rightarrow [0, 1]$ is the *true score*, $F(a) = \mathbb{E}_{z \sim \mathcal{D}}[s(a, z)]$ for a task distribution $\mathcal{D}$ and bounded per-instance score s , accessed only through noisy evaluations $\hat{F}$ and through *feedback* $y \in \mathcal{Y}$ (a scalar, a trace, an error message, a video); (iii) $q(\cdot \mid a, y, \mathcal{M})$ is the *proposer*, a fixed conditional distribution over $\mathcal{A}$ realized by a frozen LLM prompted with the current artifact, feedback, and memory; (iv) sel is a *selection rule* mapping evaluated candidates to the retained state (a single incumbent, a Pareto set, an archive, a lineage); (v) $\mathcal{M}_t$ is accumulated history. A *run* is the induced stochastic process $a_0, a_1, \dots$; for $\varepsilon \geq 0$ the *good set* is $G_\varepsilon = \{a : F(a) \geq \sup_{a'} F(a') - \varepsilon\}$, and the *hitting time* is $T = \min\{t : a_t \in G_\varepsilon\}$.

Remark 2.2 (Noise model). The true score F is bounded in $[0, 1]$, but the evaluation noise ξ in $\hat{F} = F + \xi$ is modeled as σ -subgaussian and may be unbounded (§5); this is the standard idealization for per-instance score averages and costs nothing below, since only tail bounds on ξ are used. Where boundedness of the *per-instance* score matters (§7), we use it explicitly.

The definition is deliberately minimal: the systems in Table 1 differ in engineering but not in type. Learning happens *in artifact space*; the gradient-trained model appears only as the fixed kernel q . Three structural facts drive everything that follows. First, $\mathcal{A}$ is discrete and enormous, so all guarantees must come from where q puts its mass (§3). Second, the loop’s only contact with the task is the feedback channel $\mathcal{Y}$, so its informativeness rate-limits learning (§4). Third, $\hat{F}$ is noisy and expensive, so selection is statistics, not bookkeeping (§5)—and the three constraints bind *simultaneously*, which is what §6 quantifies.

Where the inner magic lives. Our results treat q as a black-box measure. The reason a frozen LLM is a *useful* measure is the subject of the in-context-learning literature: conditioning behaves like implicit Bayesian inference (Xie et al., 2022; Akyürek et al., 2023), and trained transformers can emulate learning algorithms in context (von Oswald et al., 2023; Garg et al., 2022). We import this as

licence for the modeling assumption that feedback conditioning *increases* $q(G_\varepsilon)$ —Assumption 3.1 below—and we make the assumption falsifiable rather than free: Proposition 3.5 shows how to estimate the boost, and Appendix I specifies the experiment.

Relation to classical objects. With q fixed and $\mathcal{Y}$ trivial, Definition 2.1 is randomized program search, and Levin’s universal search (Levin, 1973; Solomonoff, 1964) with its modern descendants (Schmidhuber, 2004; Schmidhuber et al., 1997; Hutter, 2005; Achille & Soatto, 2025) applies verbatim. With $\mathcal{A}$ a population and sel an archive, it is quality-diversity evolution (Lehman & Stanley, 2011; Mouret & Clune, 2015). With sel statistical and $\hat{F}$ noisy, it is best-arm identification (Maron & Moore, 1994; Even-Dar et al., 2006; Karnin et al., 2013; Li et al., 2017). The contribution of this paper is to make the translations exact, extract the quantitative consequences for the LLM-era instantiations, and prove one statement (Theorem 6.1) that needs all the parts at once.

3 RESOURCE I: PRIOR MASS

No search rule can beat uniform enumeration on average over all problems (Wolpert & Macready, 1997); concretely, for a target placed *uniformly at random* in a space of size $|\mathcal{A}|$ —the hypothesis of Theorem 4.2(i) below—every algorithm has expected hitting time $(|\mathcal{A}| + 1)/2$. Whatever makes artifact search tractable must therefore be alignment between the proposer and the problem. The following theorem says this alignment is measured by one number.

Assumption 3.1 (Trace boost). There exist $\gamma_t \geq 0$ such that, almost surely over histories, the conditioned proposer satisfies $q(G_\varepsilon \mid a_t, y_t, \mathcal{M}_t) \geq \gamma_t q_0(G_\varepsilon)$, where q_0 is the unconditioned (cold) proposal distribution.

Theorem 3.2 (Hitting time = inverse prior mass). *Let proposals $a'_1, a'_2, \dots$ be drawn from the system’s proposer and let T be the first time $a'_t \in G_\varepsilon$.*

- (i) *If proposals are i.i.d. from q with $p := q(G_\varepsilon) > 0$, then $\mathbb{E}[T] = 1/p$, and $T \leq \ln(1/\delta)/p$ with probability $\geq 1 - \delta$.*
- (ii) *If each proposal costs compute $c(a)$ with $\bar{c} = \mathbb{E}_{a \sim q}[c(a)] < \infty$, the expected total compute is $\mathbb{E}[\sum_{t \leq T} c(a'_t)] = \bar{c}/p$.*
- (iii) *Under Assumption 3.1 with $\gamma_t \geq \gamma > 0$ for all t , $\mathbb{E}[T] \leq 1/(\gamma q_0(G_\varepsilon))$, and more generally $\mathbb{E}[T] \leq t_0 + 1/(\gamma q_0(G_\varepsilon))$ if the bound holds only for $t \geq t_0$.*

Proposition 3.3 (Lower bound). *If instead $q(G_\varepsilon \mid \cdot) \leq \bar{q}$ almost surely at every step, then $\mathbb{E}[T] \geq 1/(2\bar{q}) - 1$.*

Remark 3.4 (Misleading feedback; exact-mass version). If the conditional good-set mass *equals* $\gamma q_0(G_\varepsilon)$ at every step, then T is geometric and $\mathbb{E}[T] = 1/(\gamma q_0(G_\varepsilon))$ exactly; in particular, conditioning with $\gamma < 1$ is strictly slower than cold sampling. Under only the one-sided hypothesis of Proposition 3.3 (mass $\leq \gamma q_0(G_\varepsilon)$), the conclusion $\mathbb{E}[T] > 1/q_0(G_\varepsilon)$ follows in the regime $\gamma < 1/(2(1 + q_0(G_\varepsilon)))$, but not for all $\gamma < 1$.

Proposition 3.5 (The boost is estimable). *Suppose membership in a good set G is decidable by a verifier, and both $q_0(G) \geq q_{\min} > 0$ and $q_\tau(G) \geq q_{\min}$, where q_τ denotes the trace-conditioned proposer. Draw K i.i.d. proposals from each of q_0 and q_τ and let $\hat{\gamma} = \hat{q}_\tau(G)/\hat{q}_0(G)$ be the plug-in ratio of empirical pass rates. For any $\eta \in (0, 1]$ and $\delta \in (0, 1)$, if $K \geq \frac{9}{2q_{\min}^2 \eta^2} \ln \frac{4}{\delta}$, then with probability at least $1 - \delta$, $(1 - \eta)\gamma \leq \hat{\gamma} \leq (1 + \eta)\gamma$, where $\gamma = q_\tau(G)/q_0(G)$.*

Proofs are in Appendix A; (ii) is Wald’s identity (Wald, 1944), and the pair (i)+(iii) is an adaptive analogue of Levin’s bound, in which search cost is inverse prior mass weighted by per-program time (Levin, 1973). Proposition 3.5 is what makes Assumption 3.1 an empirical claim rather than a relabeling: the boost can be certified to constant relative error at cost $O(\ln(1/\delta)/q_{\min}^2)$ verified proposals per condition, and Appendix I turns this into a concrete protocol (sample K patch proposals from a frozen model with and without the execution trace in context, against a held-out verifier, and report the Wilson-interval ratio). Three readings matter for practice.

Pretraining buys $q_0(G_\varepsilon)$. The cold prior mass a frontier model puts on “useful edits to this artifact” is the quantity that moved between 2017-era neural program search (Zoph & Le, 2017) and FunSearch:

the search algorithm barely changed; the prior did. This is also the formal content of viewing agents as devices that convert pretraining into search-time speedup (Achille & Soatto, 2025).

Traces buy γ_t . Feedback conditioning multiplies, rather than adds to, the cold mass: a proposer that reads a stack trace and localizes the fault concentrates q on the small set of edits consistent with it. Section 4 bounds how large this multiplier can be in terms of the information carried by y . The same mechanism run backward is a warning: misleading feedback slows search below cold sampling—exactly so under the exact-mass reading, and via Proposition 3.3 once γ is small (Remark 3.4)—a regime reported when traces are uninformative for the failure mode at hand, e.g. sparse-reward exploration domains (Weng, 2026).

Reuse compounds the prior. Adding solved-task artifacts to $\mathcal{M}$ (skill libraries in Voyager, candidate pools in GEPA, lineages in DGM) re-weights q_0 for subsequent problems—the mechanism of incremental bias shift formalized for program search by Schmidhuber et al. (1997).

4 RESOURCE II: FEEDBACK INFORMATION

Fix the prior; what does richer feedback buy? We model one round as: the learner proposes $a \in \mathcal{A}$, and a feedback oracle returns $y = \phi(a, a^*) \in \mathcal{Y}$, where a^* is an unknown target artifact (the “right” prompt, patch, or program) and ϕ is the feedback protocol. Identification of a^* is the cleanest possible stand-in for “learning the task”; richer settings reduce to it locally.

Theorem 4.1 (Transcript bound). *Let $\mathcal{A}^* \subseteq \mathcal{A}$ with $|\mathcal{A}^*| = 2^L$ and let a^* be uniform on $\mathcal{A}^*$. Any (possibly randomized) algorithm that identifies a^* within T rounds with probability $\geq 1/2$, using feedback alphabet $\mathcal{Y}$ with $|\mathcal{Y}| \geq 2$, satisfies $T \geq (L - \log_2 6) / \log_2 |\mathcal{Y}|$.*

Theorem 4.2 (Exponential separation). *Let $\mathcal{A}^* = \{0, 1\}^L$ and a^* uniform.*

- (i) **Scalar success reward** ($\phi = \mathbf{1}\{a = a^*\}$): *every algorithm has $\mathbb{E}[T] \geq (2^L + 1)/2$.*
- (ii) **First-error feedback** ($\phi = \text{index of the first coordinate where } a \text{ disagrees with } a^*$): *a deterministic algorithm identifies a^* in at most L rounds.*
- (iii) **Distance feedback** ($\phi = \text{Hamming distance}$): *$L + 1$ rounds suffice.*

In (ii), “identifies” means the algorithm *knows* a^* after at most L responses; if the protocol additionally requires observing the success response itself, one confirming query is added (worst case $L + 1$). Our simulation adopts the confirming-query convention, whence the realized average $L/2 + 1$ in Figure 3.

The gap between (i) and (ii) is 2^L vs. L : *exponential*, and not an artifact of alphabet size—first-error feedback uses an alphabet of size $L+1$, only logarithmically wider as measured by Theorem 4.1, yet realizes exponentially more *usable* information because each response certifies a prefix and names a fault. This is precisely the structure of a stack trace, a failed unit test with its name, or a judge critique that says *where* the rollout went wrong. The general theory of when language feedback admits such speedups is developed by Xu et al. (2025) via the transfer eluder dimension, with benchmark instantiations in Cheng et al. (2023); Theorem 4.2 is the minimal worst-case example one can keep in one’s head, in the lineage of Shannon’s dictum that information is what reduces uncertainty (Shannon, 1948; Cover & Thomas, 2006).

Reading the empirical record through this lens: GEPA’s reported advantage over GRPO at matched rollout budgets (Agrawal et al., 2026) is the prediction of (i)-vs-(ii) for systems that read traces versus systems that collapse rollouts to scalars; the same comparison recurs as TextGrad’s textual gradients (Yuksekgonul et al., 2024), Trace’s execution-graph feedback (Cheng et al., 2024), Reflexion’s verbal self-feedback (Shinn et al., 2023), and Meta-Harness’s insistence on exposing *raw* histories rather than summaries (Lee et al., 2026). Conversely, scalar-only loops are not doomed—they are slow by exactly the measure of Theorem 4.1, which is why they appear where evaluation is verified and cheap (FunSearch’s island model makes millions of scalar-scored calls; Romera-Paredes et al., 2024).

5 SELECTION: NOISE AND DECEPTION

The selector sees neither F nor a^* ; it sees $\hat{F} = F + \xi$ under an evaluation budget. Two independent hazards arise: *statistical* (noise creates fictitious progress) and *geometric* (real progress requires

descending). Both have sharp, elementary theory; both have been rediscovered empirically at some cost.

5.1 THE RATCHET ILLUSION

Model a worst case that is also the common case at the frontier of a tuned system: a stream of *null edits*, each with true gain 0, evaluated once with σ -subgaussian noise. The greedy single-evaluation ratchet is the rule of naive prompt hill-climbing, and—as informally documented—of autoresearch (Karpathy, 2026).

Theorem 5.1 (Ratchet illusion). *Let $\hat{s}_1, \dots, \hat{s}_t$ be the measured scores of t null edits, i.i.d. σ -subgaussian, and let $m_t = \max_{i \leq t} \hat{s}_i$ be the reported best. Adopt the empty-incumbent convention: the first candidate is accepted unconditionally, and candidate i is accepted iff $\hat{s}_i > \max_{j < i} \hat{s}_j$.*

- (i) $\mathbb{E}[m_t] \leq \sigma\sqrt{2 \ln t}$, and for Gaussian noise $\mathbb{E}[m_t] \geq (1 - o(1)) \sigma\sqrt{2 \ln t}$: the reported improvement diverges although every artifact is worthless.
- (ii) The expected number of accepted edits is $H_t = \sum_{i \leq t} 1/i = \ln t + O(1)$, independent of σ . (Under a measured-incumbent convention—a baseline score $\hat{s}_0$ is drawn first and candidate i must beat the running maximum including it—the count is $H_{t+1} - 1$; the two conventions differ by $O(1)$.)

Proposition 5.2 (Cost of a sound decision). *Let evaluations be σ -subgaussian about their means.*

- (i) **Two-sided gate.** To distinguish a true gain Δ from a null edit with both error probabilities $\leq \delta$, it suffices to average $N \geq (16\sigma^2/\Delta^2) \ln(1/\delta)$ fresh evaluations of the candidate and of the incumbent and accept iff the measured gain exceeds $\Delta/2$.
- (ii) **Lower bound.** Any rule with both errors $\leq \delta$ requires $N \geq (2\sigma^2/\Delta^2) \ln(1/(4\delta))$.
- (iii) **One-sided gate.** Against an incumbent whose mean is known, $N \geq (8\sigma^2/\Delta^2) \ln(1/\delta)$ evaluations of the candidate alone suffice (accept iff the candidate’s mean exceeds the incumbent mean by $\Delta/2$).

Corollary 5.3 (Certifying a whole run). *A B -decision loop that applies Proposition 5.2 with $\delta' = \delta/B$ makes no false accept during the entire run with probability $\geq 1 - \delta$, at per-decision cost (i) $N = \lceil (16\sigma^2/\Delta^2) \ln(B/\delta) \rceil$ in the two-sided regime, or (ii) $N = \lceil (8\sigma^2/\Delta^2) \ln(B/\delta) \rceil$ against a known incumbent mean: soundness costs only a $\log B$ factor.*

Theorem 5.1 is extreme-value theory (Boucheron et al., 2013) plus the classical record-counting identity; Proposition 5.2 is a two-point testing bound (Tsybakov, 2009; Hoeffding, 1963). The practical content is blunt. A loop making $B = 2000$ single-evaluation decisions at $\sigma = \Delta$ will report meaningful-looking progress on pure noise—a realized median of 3.4σ at $t = 2000$, climbing toward the $\sigma\sqrt{2 \ln t} = 3.9\sigma$ asymptotic ceiling (Figure 2a)—and accept ≈ 8.2 null edits ($H_{2000} \approx 8.18$); the same loop gated with $N = 85$ repeats per decision (Corollary 5.3(ii)) is sound for the whole run at the 5% level (Figure 2b).

This phenomenon has an archival evidence base predating LLM loops. Expected-maximum reporting across tuning trials inflates with the number of trials exactly as Theorem 5.1(i) prescribes, which is why Dodge et al. (2019) demand expected-validation-performance curves; few-seed point estimates routinely reverse conclusions in deep RL (Henderson et al., 2018); and interval-based aggregation of precisely the Corollary 5.3 type is the remedy prescribed by Agarwal et al. (2021). The recent autoresearch episode—a circulated “53% speedup” from single 5-minute evaluations that did not survive re-evaluation, alongside an 11% gain that did (informally documented; Karpathy, 2026)—is the same theorem executed in public, and we cite it as illustration rather than evidence. Racing and successive-halving methods spend the N evaluations adaptively (Maron & Moore, 1994; Karnin et al., 2013; Li et al., 2017), and instance-aggregation designs in benchmark construction serve the same theorem (Anonymous, 2026).

5.2 DECEPTION: WHEN DESCENDING IS NECESSARY

Noise aside, greedy selection fails on landscapes where the path to the optimum passes through worse artifacts—*deception* in the evolutionary-computation sense (Lehman & Stanley, 2011; Stanley & Lehman, 2015). We exhibit the cleanest separation we know of.

Theorem 5.4 (Deception separation). *Let $\mathcal{A} = \{0, 1, \dots, 2k\}$ be a chain with*

$$f(i) = k - i \text{ on } [0, k], \quad f(i) = 2(i - k) \text{ on } [k, 2k],$$

so that $f(2k) = 2k > k = f(0)$ and the global optimum $2k$ is unique, while the slope on $[0, k]$ points away from it. Each step proposes a uniformly random element of $\{i - 1, i + 1\}$; out-of-range proposals are rejected (the walk stays). Start at 0.

(i) **Strict greedy** (accept only improvements) never reaches $2k$: it halts at 0, an artifact whose score is k below the optimum, with hitting time ∞ .

(ii) **Tolerant greedy** (accept worsening moves with probability $\varepsilon \in (0, 1)$) has $\mathbb{E}[T] \geq 2\varepsilon^{-k}$.

(iii) **Archive search** (keep all visited artifacts; expand a uniformly random archive member by a random neighbor; ignore f entirely until the end) has $\mathbb{E}[T] \leq 4k(2k + 1) = O(k^2)$.

The proof of (ii) is a birth–death drift computation; (iii) is a frontier argument (Appendix D); the acceptance dynamics, and hence the exact expectations, are unchanged by the asymmetric ascent slope, whose only role is to make the optimum unique. The theorem licenses a unification practitioners reached by convergent evolution: GEPA’s per-instance Pareto frontier keeps candidates that are best at *something* even when worse on average (Agrawal et al., 2026); ADAS retains an archive of all discovered agents (Hu et al., 2025); the DGM maintains an expanding lineage and explicitly credits open-endedness (Zhang et al., 2025a); FunSearch’s islands are archives with migration (Romera-Paredes et al., 2024). All are mechanisms for buying polynomial valley-crossing at the price of evaluating diverse, currently-worse artifacts—MAP-Elites logic (Mouret & Clune, 2015) transplanted to text space. The theorem also delimits the claim: on deception-free landscapes greedy is optimal among these rules, which is why bounded, well-gated ratchets (SkillOpt, Heuristic Learning) thrive on tasks where each failure names its own fix.

6 COMPOSITION: A CERTIFIED-IMPROVEMENT THEOREM

Each result so far isolates one resource in a deliberately minimal sub-model. But Definition 2.1 places the resources in one object, and a fair criticism of any dictionary is that nothing yet *requires* the assembled object. The following theorem does. It runs the §3 proposer against the §5.1 evaluator under the Corollary 5.3 gate, and it accounts for the interaction the components inflict on each other: the gate that blocks fictitious progress also occasionally rejects true improvements, slowing the chain—the cost appears explicitly as the factor $(1 - \delta/B)$.

Call $F(\Delta, \varepsilon)$ -separated if $F(a) - F(a') \geq \Delta$ for every $a \in G_\varepsilon$ and every $a' \notin G_\varepsilon$: no artifact scores within Δ below the ε -good level. This is the margin that makes finite-sample certification possible at all (Proposition 5.2(ii)).

Theorem 6.1 (Certified improvement at the compound rate). *Consider a run of the loop of Definition 2.1 in which: the proposer satisfies Assumption 3.1 with $\gamma_t \geq \gamma > 0$ and $p_\gamma := \gamma q_0(G_\varepsilon) > 0$; the evaluator is σ -subgaussian with noise independent across evaluations; selection is the incumbent ratchet in which every candidate is gated by the two-sided test of Proposition 5.2(i) at confidence $\delta' = \delta/B$, i.e. with $N = \lceil (16\sigma^2/\Delta^2) \ln(B/\delta) \rceil$ fresh evaluations per side; F is (Δ, ε) -separated; $a_0 \notin G_\varepsilon$; and $\delta \in (0, \frac{1}{2}]$ with proposal budget $B \geq 2 \ln(1/\delta)/p_\gamma$. Then with probability at least $1 - 2\delta$, simultaneously:*

(i) **Soundness**: no candidate whose true score does not exceed its incumbent’s is ever accepted;

(ii) **Liveness**: an artifact of G_ε is proposed, certified, and accepted within $T \leq \lceil 2 \ln(1/\delta)/p_\gamma \rceil$ gated decisions; moreover $\mathbb{E}[T] \leq 1/(p_\gamma(1 - \delta/B))$;

(iii) **Cost**: the total number of evaluator calls up to that acceptance is at most

$$2NT = O\left(\frac{\sigma^2}{\Delta^2} \cdot \frac{\ln(B/\delta) \ln(1/\delta)}{\gamma q_0(G_\varepsilon)}\right).$$

The proof (Appendix E) is short because the parts were built to fit: the gate’s two error directions come from the same Hoeffding bound, separation converts “ ε -good candidate against a not-yet- ε -good incumbent” into “true gain $\geq \Delta$,” and the geometric thinning of Theorem 3.2(iii) survives because Assumption 3.1 is stated almost surely over histories—which, in a gated run, include the gate outcomes. Three remarks locate the statement.

It needs the whole object. Delete the gate and (i) fails at rate $\ln t$ (Theorem 5.1); delete the prior ($p_\gamma \rightarrow 0$) and (ii) blows up (Proposition 3.3); delete the noise ($\sigma = 0$) and the statement collapses to Theorem 3.2. The three resources of §3 and §5.1 multiply in (iii), which is the quantitative sense in which the loop is one object and not five literatures stapled together.

False rejects are a rate, not a failure. The gate rejects a true Δ -improvement with probability at most δ/B ; the chain pays the factor $(1 - \delta/B) \geq \frac{1}{2}$ in (ii) and nothing else. Certification is compatible with liveness at the stated budget—“soundness costs only log factors” extends from single decisions (Corollary 5.3) to the full search.

The margin is honest. (Δ, ε) -separation is an assumption about the landscape, not a free lunch; without a margin, no finite-sample gate can distinguish arbitrarily small true gains from noise (Proposition 5.2(ii)). The one-sided variant (gate (iii) of Proposition 5.2) halves the constant when incumbents are re-certified rather than re-measured.

7 GENERALIZATION: THE ARTIFACT AS COMPRESSION

What is the evolved artifact, statistically? It is chosen *because* it scored well on n validation instances; its value is its score on fresh ones. Since artifacts are strings, the right uniform-convergence currency is description length, in the MDL tradition (Rissanen, 1978) and its PAC-Bayes form (McAllester, 1999).

Theorem 7.1 (Occam bound for artifact selection). *Fix a prefix-free code with lengths $\ell(a)$ (bits) on $\mathcal{A}$, let $\hat{F}_n(a)$ be the mean score on n i.i.d. validation instances, and let $\delta \in (0, 1)$. With probability $\geq 1 - \delta$, simultaneously for all $a \in \mathcal{A}$ —hence for the selected $\hat{a}$, no matter how adaptively chosen—*

$$|F(a) - \hat{F}_n(a)| \leq \sqrt{\frac{\ell(a) \ln 2 + \ln(2/\delta)}{2n}}.$$

Corollary 7.2 (The U-shape: collapse vs. bloat). *Let $A(L) = \sup_a F(a) - \max_{\ell(a) \leq L} F(a)$ be the approximation error of length- L artifacts and let $\hat{a}_L$ maximize $\hat{F}_n$ over $\{\ell(a) \leq L\}$. Then with probability $\geq 1 - \delta$,*

$$\sup_a F(a) - F(\hat{a}_L) \leq A(L) + 2\sqrt{\frac{L \ln 2 + \ln(2/\delta)}{2n}}.$$

The two terms move in opposite directions in L , and the corollary is a quantitative statement of the two documented failure modes of context evolution (Zhang et al., 2025b): *context collapse* is operating at L below the knee of $A(L)$ —the artifact is too short to encode the task statistic it is supposed to carry—while *validation overfitting* is letting L grow with n fixed, the regime in which GEPA-style optimizers are observed to degrade on held-out instances. It also derives, rather than assumes, two disciplines that the strongest gated systems adopted empirically: SkillOpt’s bounded add/delete/replace edits control the growth of $\ell(a)$ per accepted step (Yang et al., 2026), and Heuristic Learning’s instruction to “absorb feedback and compress history” (Weng, 2026) is explicit description-length management; memory-curation systems live in the same trade-off (Packer et al., 2023; Suzgun et al., 2025). The bound’s $\sqrt{\ell/n}$ scaling makes a falsifiable prediction: validation set size must grow linearly in artifact bits to keep the certified gap constant. We are not aware of a published $(\ell(a), n, \text{gap})$ measurement on a deployed optimizer; §11 states the experiment.

8 LIMITS: PROXY GAPS AND SELF-REFERENCE

Every guarantee so far conditions on the evaluator measuring what we mean. Let $\tilde{F} = F + b$ be the proxy actually optimized, with bias b (judge error, reward misspecification, test-set leakage).

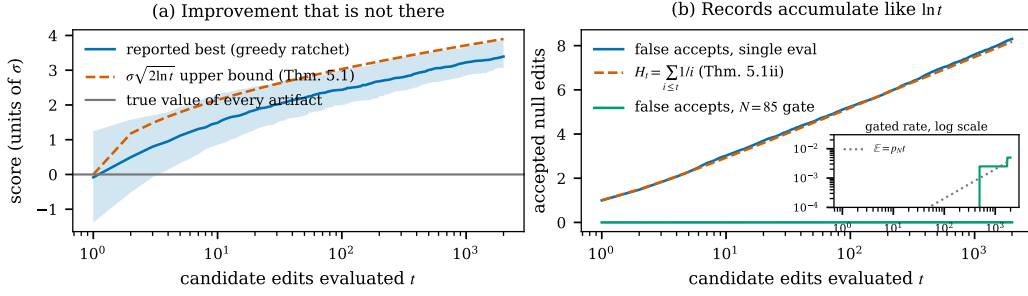


Figure 2: The ratchet illusion (Theorem 5.1, Corollary 5.3). Null edits, $\sigma = 1$, 400 runs, seed committed. **(a)** Reported best score of a single-evaluation greedy ratchet (median, 10–90% band) against the $\sigma\sqrt{2\ln t}$ upper bound; the realized median at $t = 2000$ is 3.39σ (mean 3.44σ) against the bound’s 3.90σ —the visible gap is the second-order extreme-value correction. The true value of every artifact is 0. **(b)** Accepted null edits track H_t (simulated 8.30 vs. $H_{2000} = 8.18$); gating each decision with $N = 85$ repeats (Corollary 5.3(ii) at $B = 2000$, $\delta = 0.05$, $\Delta = \sigma$) yields 2 false accepts across the 8×10^5 gated decisions—a per-run rate of 0.005, within the $\delta = 5\%$ certificate and consistent with the Poisson prediction 1.61 (inset, log scale, against $\mathbb{E} = p_N t$ with $p_N \approx 2 \times 10^{-6}$).

Proposition 8.1 (Goodhart transfer and its failure). *(i) **Bounded bias transfers.** If $|b(a)| \leq \alpha$ on all reachable artifacts and a run certifies measured progress $\tilde{F}(a_T) - \tilde{F}(a_0) \geq g$, then $F(a_T) - F(a_0) \geq g - 2\alpha$.*

*(ii) **Selection harvests bias.** If $\hat{a}$ maximizes $\tilde{F}$ over m candidates whose biases are independent σ_b -subgaussian, the harvested bias satisfies $\mathbb{E}[b(\hat{a})] \leq \sigma_b\sqrt{2\ln m}$, and this rate is attained for Gaussian bias on otherwise-equal candidates.*

*(iii) **Evaluator integrity is necessary.** If the loop can modify the map $a \mapsto b(a)$ along reachable paths (an editable evaluator), there exist systems within the model ($F : \mathcal{A} \rightarrow [0, 1]$) in which the proxy ratchet accepts every step, measured score increases strictly with $\tilde{F}(a_t) \uparrow 1$, and true score decreases strictly with $F(a_t) \downarrow 0$ from $F(a_0) = 1$: measured progress stays positive while true progress traverses the entire range downward. Consequently no transfer inequality $F\text{-progress} \geq \psi(\tilde{F}\text{-progress})$ with ψ exceeding the vacuous bound -1 on positive arguments can hold.*

Part (ii) is the optimizer’s curse expressed for artifact search: optimization pressure converts evaluator variance into systematic overestimate at a $\sqrt{2\ln m}$ rate—*regressional* Goodhart in the taxonomy of Manheim & Garrabrant (2018)—which compounds the noise effects of Theorem 5.1. Part (iii) is the *adversarial* corner: its hypothesis sounds exotic until one recalls documented instances—agents hallucinating tool logs to satisfy checks in the DGM (Zhang et al., 2025a), evolved reward functions exploiting simulator physics (Ma et al., 2024)—all special cases of reward hacking (Amodei et al., 2016). The constructive reading: freeze the evaluator, hold it out from the proposer’s edit scope, and re-estimate bias on fresh data, and (i)+(ii) bound the damage; let the loop touch it and the guarantee is not weakened but *void*. The self-referential limit—improvers improving improvers (Schmidhuber, 2003; Zelikman et al., 2024; Fernando et al., 2023)—inherits exactly this condition, and we record in Appendix J the open problem of rates (Yampolskiy, 2015; Chalmers, 2010).

9 NUMERICAL ILLUSTRATIONS

We illustrate the three quantitative theorem families in controlled simulation; the seeded code is in the supplement and all figures regenerate in under a minute on a laptop. Every arm in every figure is an *executed algorithm*; exact laws are plotted as separate, labeled curves. These are illustrations of exact statements, not benchmark claims.

Figure 2 shows the single-evaluation ratchet manufacturing a median 3.4σ “improvement” from 2000 worthless edits, with false accepts tracking H_t ; the $N=85$ gate of Corollary 5.3(ii) reduces false

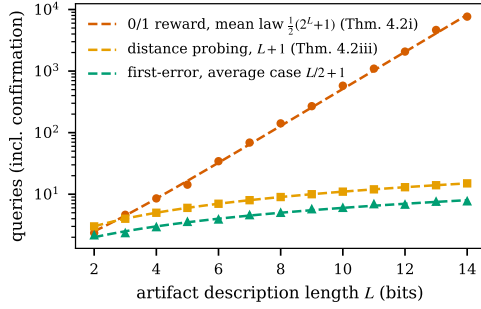


Figure 3: Feedback separation (Theorems 4.1–4.2). Mean queries to identify an L -bit target under three feedback protocols, *each executed as an algorithm* (60 runs; markers), against exact laws (dashed): the scalar arm’s mean hitting law $(2^L+1)/2$; the distance probe’s deterministic $L+1$; and the first-error arm’s *average case* $L/2 + 1$ under the confirming-query convention (Theorem 4.2(ii) bounds identification by L). The scalar-vs-structured gap is exponential.

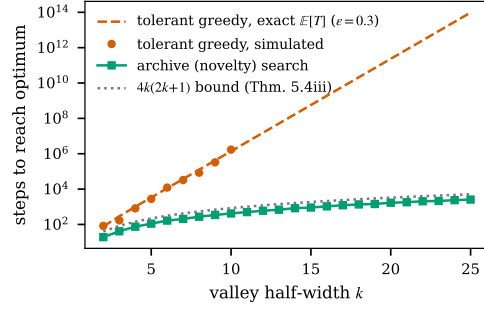


Figure 4: Deception separation (Theorem 5.4) on the unique-optimum landscape ($f(2k) = 2k > k = f(0)$). Steps to cross a deceptive valley of half-width k : tolerant greedy ($\epsilon = 0.3$; exact expectation and simulation) grows exponentially; archive search stays under the $4k(2k+1)$ bound. Strict greedy never arrives (not plottable).

accepts to the certified trickle shown honestly in the inset. (An earlier draft of this paper reported the gated count as “zero” from this single seed; the theory’s own Poisson prediction is ≈ 1.6 , and replaying the seed finds 2. We keep the correction visible as a one-seed illusion caught by the paper’s own theorem; see Appendix H.) Figure 3 realizes the exponential-vs-linear separation of Theorem 4.2 with all three protocols executed end to end. Figure 4 shows tolerant greedy’s exponential wall against the archive’s quadratic crossing, including the exact birth–death expectation. Appendix H contains parameters, conventions, and verification printouts.

10 RELATED WORK

Empirical beyond-gradient systems. Prompt optimization spans APE, OPRO, PromptBreeder, and program-level compilers (Zhou et al., 2023; Yang et al., 2024; Fernando et al., 2023; Khattab et al., 2024; Opsahl-Ong et al., 2024), culminating in trace-reflective methods (Agrawal et al., 2026; Yuksekgonul et al., 2024; Cheng et al., 2024). Context and memory evolution includes ACE, MemGPT, and test-time memory (Zhang et al., 2025b; Packer et al., 2023; Suzgun et al., 2025); skills and heuristics, Yang et al. (2026) and Weng (2026); agent/harness search, Hu et al. (2025), Zhang et al. (2025a), and Lee et al. (2026); code and discovery loops, Romera-Paredes et al. (2024), Novikov et al. (2025), and Karpathy (2026); self-referential variants, Zelikman et al. (2024). We add no system; we add the common load-bearing mathematics.

Evaluation statistics. The selection results of §5.1 are the artifact-search reading of an archival literature on fictitious improvement under noisy evaluation: seed sensitivity and point-estimate fragility in deep RL (Henderson et al., 2018), expected-max inflation across tuning trials (Dodge et al., 2019), and interval-based aggregate reporting (Agarwal et al., 2021). Our contribution is the closed-form ratchet law, the matching certification cost, and their composition with search (Theorem 6.1).

Theory. Our prior-mass results adapt universal search (Levin, 1973; Schmidhuber, 2004; Hutter, 2005) and its recent agentic reading (Achille & Soatto, 2025). The feedback results are worst-case companions to learning from language feedback via transfer eluder dimension (Xu et al., 2025; Cheng et al., 2023), against the scalar-sufficiency position of Silver et al. (2021): our account is quantitative rather than oppositional—scalars suffice in the limit, traces change the *rate*. Selection-under-noise imports best-arm identification (Even-Dar et al., 2006; Karnin et al., 2013; Li et al., 2017); deception imports quality-diversity theory (Lehman & Stanley, 2011; Mouret & Clune, 2015) and runtime analysis of evolutionary algorithms (Droste et al., 2002); generalization imports MDL/PAC-Bayes

(Rissanen, 1978; McAllester, 1999); limits import No Free Lunch (Wolpert & Macready, 1997) and Goodhart taxonomies (Manheim & Garrabrant, 2018; Amodei et al., 2016). Methodologically we follow the settle-the-question genre of Bowling et al. (2023) and Richens et al. (2025).

11 DISCUSSION AND CONCLUSION

One loop, five dials, one certificate. The model of Definition 2.1 reduces a crowded design space to five measurable quantities—cold prior mass $q_0(G_\varepsilon)$, trace boost γ , feedback information per rollout, noise-gap ratio σ^2/Δ^2 , and artifact description length $\ell(a)$ —plus one integrity condition on the evaluator; and Theorem 6.1 states how the statistical dials multiply into a single certified cost. Each result is a falsifiable scaling claim: hitting time should scale inversely with measured prior mass (testable by ablating proposer quality); the trace boost γ is directly estimable at known sample cost (Proposition 3.5; protocol in Appendix I, with the buggy-program experiment specified to the level of prompts and intervals); scalar-vs-trace gaps should scale with the feedback’s usable bits (testable by degrading traces); required repeats should scale as $(\sigma^2/\Delta^2) \log B$ (testable in any ratchet); and certified generalization gaps should scale as $\sqrt{\ell(a)/n}$ —for which the cleanest missing measurement is a $(\ell(a), n, \text{val-test gap})$ curve from a deployed prompt optimizer, stratified by evolved-prompt length.

Limitations. Our model treats the proposer as a fixed measure and the task as stationary; it does not explain *why* LLM priors align with useful edits (the ICL literature’s question), nor handle non-stationary evaluators, multi-objective artifacts, or interaction between concurrent loops. The deception and feedback constructions are minimal worst cases: they bound what is possible, not what is typical. The mapping from theorems to deployed-system behavior remains qualitative: we have grounded the selection results in archival measurements from neighboring literatures and flagged informal sources explicitly, but no experiment in this paper measures the five dials on a live LLM loop; Appendix I specifies the first such measurement, and running it is the natural next step, as is the open self-reference problem of Appendix J.

Conclusion. Learning beyond gradients is not a new kind of learning awaiting a new mathematics. It is stochastic search over program space with a learned prior, a wide feedback channel, statistical selection, and description-length generalization—five old literatures meeting one new proposal distribution, and composing, when gated honestly, into certified improvement at a compound rate. Naming the correspondence buys the field what theory is for: its successes become predictions, its failures become theorems, and its next systems can be designed against bounds rather than anecdotes.

ETHICS STATEMENT

This paper advances the theoretical understanding of self-improving LLM systems. Sharper theory for when such loops are sound (Corollary 5.3, Theorem 6.1, Proposition 8.1) primarily supports safer engineering: it gives auditable conditions under which claimed autonomous improvements are real, and bounds on proxy gaming. The same analysis could marginally inform attempts to build more capable self-improving systems; we judge the safety-relevant clarity—particularly the evaluator-integrity condition—to outweigh this, and we discuss failure modes throughout rather than in isolation.

REPRODUCIBILITY STATEMENT

All proofs are self-contained in Appendices A–G with explicit constants. All simulations are pure NumPy, seeded, and complete in under a minute on one CPU core (Appendix H); the committed script regenerates every figure and prints every numeric claim made in §5.1 and §9 (realized median 3.39σ ; $H_{2000} = 8.18$ vs. simulated 8.30; 2 gated false accepts against a Poisson prediction of 1.61), so prose and artifacts can be audited against each other—an audit this revision itself benefited from. The γ -measurement protocol of Appendix I is specified to the level of prompts, sample sizes, and confidence intervals.

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A PROOFS FOR SECTION 3 (PRIOR MASS)

Proof of Theorem 3.2. (i) With i.i.d. proposals, T is geometric with success probability p : $\mathbb{P}(T > t) = (1-p)^t$, so $\mathbb{E}[T] = \sum_{t \geq 0} (1-p)^t = 1/p$. For the tail, $(1-p)^t \leq e^{-pt} \leq \delta$ once $t \geq \ln(1/\delta)/p$.

(ii) T is a stopping time for the i.i.d. sequence (a'_t) : the event $\{T \geq t\} = \{a'_1 \notin G_\varepsilon, \dots, a'_{t-1} \notin G_\varepsilon\}$ is determined by the first $t-1$ proposals, hence independent of $c(a'_t)$. Since $\mathbb{E}[T] = 1/p < \infty$ and $\mathbb{E}[c(a'_1)] = \bar{c} < \infty$ with $c \geq 0$, Wald's identity (Wald, 1944) gives $\mathbb{E}[\sum_{t=1}^T c(a'_t)] = \mathbb{E}[T] \bar{c} = \bar{c}/p$.

(iii) Let $p_t = q(G_\varepsilon \mid a_t, y_t, \mathcal{M}_t)$ be the conditional hit probability at step t , so $p_t \geq \gamma q_0(G_\varepsilon) =: p_\gamma$ almost surely. Then

$$\mathbb{P}(T > t) = \mathbb{E}\left[\prod_{i=1}^t (1-p_i)\right] \leq (1-p_\gamma)^t,$$

by iterated conditioning (peel off the last factor: $\mathbb{E}[\prod_{i \leq t} (1-p_i)] = \mathbb{E}[\prod_{i \leq t-1} (1-p_i) \mathbb{E}[(1-p_t) \mid \mathcal{F}_{t-1}]] \leq (1-p_\gamma) \mathbb{E}[\prod_{i \leq t-1} (1-p_i)]$, and induct). Summing the tail gives $\mathbb{E}[T] \leq 1/p_\gamma$. If the boost holds only after t_0 , bound the first t_0 steps by t_0 and apply the argument to the remainder. $\square$

Proof of Proposition 3.3. By the union bound and iterated conditioning, $\mathbb{P}(T \leq t) \leq \sum_{i=1}^t \mathbb{E}[p_i] \leq t\bar{q}$. Hence, with $m = \lfloor 1/\bar{q} \rfloor$,

$$\mathbb{E}[T] = \sum_{t \geq 0} \mathbb{P}(T > t) \geq \sum_{t=0}^{m-1} (1-t\bar{q}) = m - \bar{q} \frac{m(m-1)}{2} \geq m - \frac{m}{2} = \frac{m}{2} \geq \frac{1}{2\bar{q}} - 1.$$

(The second inequality uses $\bar{q}(m-1) \leq 1$.) $\square$

Proof of Remark 3.4. If the conditional mass equals $\gamma q_0(G_\varepsilon)$ at every step, the indicator of a hit at each step is Bernoulli($\gamma q_0(G_\varepsilon)$) conditionally on the past, so $\mathbb{P}(T > t) = (1 - \gamma q_0(G_\varepsilon))^t$ exactly and $\mathbb{E}[T] = 1/(\gamma q_0(G_\varepsilon))$, which exceeds the cold rate $1/q_0(G_\varepsilon)$ iff $\gamma < 1$. Under the one-sided hypothesis with $\bar{q} = \gamma q_0(G_\varepsilon)$, Proposition 3.3 gives $\mathbb{E}[T] \geq 1/(2\gamma q_0(G_\varepsilon)) - 1$, which exceeds $1/q_0(G_\varepsilon)$ iff $\gamma < 1/(2(1 + q_0(G_\varepsilon)))$; for γ between this threshold and 1 the one-sided hypothesis alone is silent, which is why the prose claim in §3 is stated through the exact-mass version. $\square$

Proof of Proposition 3.5. Write $q_0 = q_0(G)$, $q_\tau = q_\tau(G)$, and $e_K = \sqrt{\ln(4/\delta)/(2K)}$. Each empirical pass rate is a mean of K i.i.d. Bernoulli variables, so Hoeffding's inequality (Hoeffding, 1963) gives $\mathbb{P}(|\hat{q}_0 - q_0| \geq e_K) \leq 2e^{-2Ke_K^2} = \delta/2$ and likewise for $\hat{q}_\tau$; on the intersection (probability $\geq 1 - \delta$) both estimates are within e_K of their means. The sample-size condition is exactly $e_K \leq \eta q_{\min}/3$; write $x = \eta/3 \leq 1/3$. Since $q_0, q_\tau \geq q_{\min}$, we have $e_K/q_0 \leq x$ and $e_K/q_\tau \leq x$, and $\hat{q}_0 \geq q_0 - e_K \geq (1-x)q_{\min} > 0$, so $\hat{\gamma}$ is well defined. Then

$$\hat{\gamma} \leq \frac{q_\tau + e_K}{q_0 - e_K} = \gamma \frac{1 + e_K/q_\tau}{1 - e_K/q_0} \leq \gamma \frac{1+x}{1-x} \leq \gamma(1+3x) = \gamma(1+\eta),$$

using $(1+x)/(1-x) \leq 1+3x$ for $x \leq 1/3$; and symmetrically $\hat{\gamma} \geq \gamma(1-x)/(1+x) \geq \gamma(1-2x) \geq \gamma(1-\eta)$. $\square$

B PROOFS FOR SECTION 4 (FEEDBACK INFORMATION)

Proof of Theorem 4.1. First take the algorithm deterministic. Its behavior is a decision tree: the query at round $t+1$ is a fixed function of the transcript $(y_1, \dots, y_t) \in \mathcal{Y}^t$, and the final output is a fixed function of $(y_1, \dots, y_T)$. Run the algorithm against every $a^* \in \mathcal{A}^*$. The number of distinct realizable transcripts of length T is at most $|\mathcal{Y}|^T$, so the final-output map takes at most $|\mathcal{Y}|^T$ distinct values. Similarly, the query asked at round $t+1$ is determined by a transcript prefix in $\mathcal{Y}^t$, so over all rounds and all runs the set of artifacts ever queried has size at most $\sum_{t=0}^{T-1} |\mathcal{Y}|^t \leq 2|\mathcal{Y}|^T$ for $|\mathcal{Y}| \geq 2$.

Credit the algorithm with success if it ever queries a^* or outputs a^* (this is generous). The set S of targets on which it can succeed therefore satisfies $|S| \leq |\mathcal{Y}|^T + 2|\mathcal{Y}|^T = 3|\mathcal{Y}|^T$. For uniform a^* ,

$$\frac{1}{2} \leq \mathbb{P}(\text{success}) = \frac{|S|}{2^L} \leq \frac{3|\mathcal{Y}|^T}{2^L} \implies |\mathcal{Y}|^T \geq \frac{2^L}{6} \implies T \geq \frac{L - \log_2 6}{\log_2 |\mathcal{Y}|}.$$

A randomized algorithm is a mixture over deterministic ones (condition on the random seed); since the success probability is the mixture average, some deterministic realization succeeds with probability $\geq 1/2$, and the bound applies to it. (Fano-type arguments of this form are standard; see Cover & Thomas, 2006.) $\square$

Proof of Theorem 4.2. (i) Until the needle is hit, every response equals 0, so a deterministic algorithm's queries form a fixed sequence $b_1, b_2, \dots$ (repetitions only hurt; assume distinct). The hitting round is the position of a^* in this sequence. For uniform a^* over 2^L artifacts, $\mathbb{E}[T] = \sum_{j=1}^{2^L} j \cdot 2^{-L} = (2^L + 1)/2$ when the sequence enumerates $\mathcal{A}^*$, and is larger otherwise. Averaging over the seed extends this to randomized algorithms.

(ii) Maintain a current string a , initialized arbitrarily. Query a ; if correct, done; otherwise the feedback names the first index i where $a_i \neq a_i^*$. Flip a_i . This flip makes coordinate i correct and does not alter coordinates $j < i$, which were already correct (the first error was at i). Hence the length of the correct prefix increases strictly with each query, and at most L queries are needed to know a^* ; a further confirming query observes the success response if the protocol requires it.

(iii) Query a base string a^0 and record $d_0 = d_H(a^0, a^*)$. For $i = 1, \dots, L$ query $a^0 \oplus e_i$: the distance is $d_0 - 1$ if coordinate i of a^0 is wrong and $d_0 + 1$ if it is right. After $L + 1$ queries every coordinate's correctness is known, determining a^* exactly. $\square$

C PROOFS FOR SECTION 5 (NOISE)

Proof of Theorem 5.1. (i) **Upper bound.** For any $\lambda > 0$, by Jensen's inequality and the subgaussian moment-generating bound $\mathbb{E}[e^{\lambda \hat{s}_i}] \leq e^{\lambda^2 \sigma^2 / 2}$,

$$e^{\lambda \mathbb{E}[m_t]} \leq \mathbb{E}[e^{\lambda m_t}] \leq \sum_{i=1}^t \mathbb{E}[e^{\lambda \hat{s}_i}] \leq t e^{\lambda^2 \sigma^2 / 2} \implies \mathbb{E}[m_t] \leq \frac{\ln t}{\lambda} + \frac{\lambda \sigma^2}{2},$$

and choosing $\lambda = \sqrt{2 \ln t} / \sigma$ yields $\mathbb{E}[m_t] \leq \sigma \sqrt{2 \ln t}$.

(i) **Lower bound (Gaussian case).** Fix $\eta \in (0, 2)$ and set $x_t = \sqrt{(2 - \eta) \ln t}$. Using the Gaussian tail bound $1 - \Phi(x) \geq \frac{x}{1+x^2} \varphi(x)$,

$$t(1 - \Phi(x_t)) \geq t \cdot \frac{x_t}{1+x_t^2} \cdot \frac{e^{-x_t^2/2}}{\sqrt{2\pi}} = \frac{x_t}{(1+x_t^2)\sqrt{2\pi}} t^{\eta/2} \xrightarrow{t \rightarrow \infty} \infty.$$

Hence $\mathbb{P}(m_t/\sigma \leq x_t) = \Phi(x_t)^t \leq \exp\{-t(1 - \Phi(x_t))\} \rightarrow 0$. Since $m_t \geq \hat{s}_1$ pointwise, $\mathbb{E}[m_t \mathbf{1}\{m_t \leq \sigma x_t\}] \geq -\mathbb{E}[\hat{s}_1] \geq -\sigma \sqrt{2/\pi}$, so

$$\mathbb{E}[m_t] \geq \sigma x_t \mathbb{P}(m_t > \sigma x_t) - \sigma \sqrt{2/\pi} = (1 - o(1)) \sigma \sqrt{(2 - \eta) \ln t}.$$

Letting $\eta \downarrow 0$ along $t \rightarrow \infty$ gives the claim. The second-order correction is negative—for Gaussian maxima the median sits $\approx (\ln \ln t + \ln 4\pi)/(2\sqrt{2 \ln t})$ below $\sigma \sqrt{2 \ln t}$ (Boucheron et al., 2013)—which is the visible gap in Figure 2a and the reason the realized $t = 2000$ median is 3.39σ rather than 3.90σ .

(ii) Under the empty-incumbent convention, the i -th candidate is accepted iff $\hat{s}_i > \max_{j < i} \hat{s}_j$, i.e. iff $\hat{s}_i$ is a *record*. For i.i.d. draws from a continuous distribution, each of the first i values is equally likely to be the largest, so $\mathbb{P}(\hat{s}_i \text{ is a record}) = 1/i$, and linearity of expectation gives $\mathbb{E}[\#\text{accepts up to } t] = H_t$. Under the measured-incumbent convention the relevant sequence is $(\hat{s}_0, \hat{s}_1, \dots, \hat{s}_t)$ and candidate $i \geq 1$ is accepted iff it is a record of that extended sequence, with probability $1/(i+1)$; summing gives $H_{t+1} - 1$. (With atoms, ties only reduce records.) $\square$

Proof of Proposition 5.2. (i) **Two-sided gate.** Estimate each side by an average of N fresh evaluations and let $\hat{D}$ be the difference; $\hat{D} - D$ is a sum of $2N$ independent σ -subgaussian terms scaled by $1/N$, hence $\sqrt{2/N} \sigma$ -subgaussian. The rule “accept iff $\hat{D} \geq \Delta/2$ ” errs only if $|\hat{D} - D| \geq \Delta/2$ (under either hypothesis $D = 0$ or $D = \Delta$), which has probability at most $\exp\{-(\Delta/2)^2/(2 \cdot 2\sigma^2/N)\} = \exp\{-N\Delta^2/(16\sigma^2)\} \leq \delta$ for $N \geq (16\sigma^2/\Delta^2) \ln(1/\delta)$.

(ii) Lower bound. A decision rule with both errors $\leq \delta$ is a test between $P_0 = \mathcal{N}(0, \sigma^2)^{\otimes N}$ and $P_1 = \mathcal{N}(\Delta, \sigma^2)^{\otimes N}$ (the hardest noise instance consistent with the model). By the Bretagnolle–Huber inequality (Tsybakov, 2009), the sum of the two error probabilities of any test satisfies

$$2\delta \geq \frac{1}{2} e^{-\text{KL}(P_0, P_1)} = \frac{1}{2} e^{-N\Delta^2/(2\sigma^2)} \implies N \geq \frac{2\sigma^2}{\Delta^2} \ln \frac{1}{4\delta},$$

nonvacuous for $\delta < 1/4$. (The constant $\ln(1/(4\delta))$ is what the argument yields; an earlier version of this paper stated $\ln(1/(2\delta))$, which is stronger than proven.)

(iii) One-sided gate. The candidate’s N -evaluation mean M satisfies: $M - \mu_{\text{cand}}$ is $\sigma/\sqrt{N}$ -subgaussian. Accept iff $M \geq \mu_{\text{inc}} + \Delta/2$, where μ_{inc} is the known incumbent mean. If the true gain is ≤ 0 , acceptance requires $M - \mu_{\text{cand}} \geq \Delta/2$; if the true gain is $\geq \Delta$, rejection requires $\mu_{\text{cand}} - M \geq \Delta/2$. Either event has probability at most $\exp\{-(\Delta/2)^2/(2\sigma^2/N)\} = \exp\{-N\Delta^2/(8\sigma^2)\} \leq \delta$ for $N \geq (8\sigma^2/\Delta^2) \ln(1/\delta)$. $\square$

Proof of Corollary 5.3. Apply Proposition 5.2(i) (resp. (iii)) with confidence $\delta' = \delta/B$ to each of the B decisions and take a union bound over the run. The per-decision cost is $N = \lceil (16\sigma^2/\Delta^2) \ln(B/\delta) \rceil$ (resp. $\lceil (8\sigma^2/\Delta^2) \ln(B/\delta) \rceil$). $\square$

D PROOFS FOR SECTION 5 (DECEPTION)

Proof of Theorem 5.4. **(i)** At state 0, the proposal is -1 (rejected as out of range; the walk stays) or 1, and $f(1) = k - 1 < k = f(0)$, so a strict-improvement rule rejects it; the chain remains at 0 forever, and $f(0) = k < 2k = f(2k)$. (From any start in $[1, k]$, only leftward moves are improvements, so the walk descends to 0 and then halts.)

(ii) In the valley interior $i \in [1, k - 1]$, a step proposes left or right with probability $1/2$ each; the left move is improving (always accepted), the right move is worsening (accepted with probability ε). Thus the per-step transition probabilities are $p = \varepsilon/2$ (right), $q = 1/2$ (left), and stay otherwise. At $i = 0$, $p = \varepsilon/2$ and $q = 0$ (the left proposal is rejected as out of range). On the ascent $i \in (k, 2k)$, the right move is improving since f is increasing there ($p = 1/2$) and the left move worsens ($q = \varepsilon/2$); at the floor $i = k$ both neighbors improve ($p = q = 1/2$). Note these probabilities—and hence all hitting times—are identical to those of the symmetric-slope landscape: the asymmetric ascent changes f -values, not accept/reject comparisons. Let h_i be the expected time to first reach $i + 1$ from i . First-step analysis for $i \geq 1$ in the valley:

$$h_i = 1 + q(h_{i-1} + h_i) + (1 - p - q)h_i \implies h_i = \frac{1 + qh_{i-1}}{p} = \frac{2}{\varepsilon} + \frac{1}{\varepsilon}h_{i-1}, \quad h_0 = \frac{2}{\varepsilon},$$

where $h_0 = 2/\varepsilon$ reflects the stated boundary convention (the proposal at 0 is right with probability $1/2$ and accepted with probability ε). By induction $h_i \geq \varepsilon^{-i}h_0 = 2\varepsilon^{-(i+1)}$. The time to reach $2k$ from 0 is at least the time to first reach k , which is $\sum_{i=0}^{k-1} h_i \geq h_{k-1} \geq 2\varepsilon^{-k}$.

(iii) Let $r_t \leq 2k$ be the rightmost element of the archive at step t ($r_0 = 0$), and note the archive size never exceeds $2k + 1$. While $r_t < 2k$, the step extends the frontier ($r_{t+1} = r_t + 1$) whenever the sampler picks the archive element r_t (probability $\geq 1/(2k + 1)$) and proposes its right neighbor (probability $1/2$; $r_t + 1$ is not in the archive by maximality of r_t). Hence each frontier advance is stochastically dominated by a geometric random variable with success probability $1/(2(2k + 1))$, and $2k$ advances are needed:

$$\mathbb{E}[T] \leq 2k \cdot 2(2k + 1) = 4k(2k + 1).$$

The rule never consults f , so deception is irrelevant to it; it pays instead in evaluations of artifacts that are currently worse, which is the quantitative trade championed by novelty search (Lehman & Stanley, 2011). $\square$

Remark D.1. The constants in (ii) are not optimized; any acceptance rule whose probability of accepting a worsening move is at most ε obeys the same bound, including Metropolis rules at fixed temperature (Kirkpatrick et al., 1983). Birth–death hitting-time formulas are classical (Levin & Peres, 2017). The exact expectation plotted in Figure 4 sums h_i over the full chain including the ascent segment.

E PROOF FOR SECTION 6 (COMPOSITION)

Proof of Theorem 6.1. Throughout, a *decision* at step t consists of drawing a candidate $a'_t \sim q(\cdot \mid a_t, y_t, \mathcal{M}_t)$, drawing N fresh evaluations of a'_t and N of the incumbent a_t (evaluation noise independent of everything previous), and accepting iff the measured gain is $\geq \Delta/2$. Write $D_t = F(a'_t) - F(a_t)$ for the true gain and let $\mathcal{F}_{t-1}$ denote the history up to and including decision $t-1$ (proposals, evaluations, gate outcomes).

Gate error bounds. Conditioning on (a'_t, a_t) , the proof of Proposition 5.2(i) gives, for the chosen $N = \lceil (16\sigma^2/\Delta^2) \ln(B/\delta) \rceil$:

$$\mathbb{P}(\text{accept} \mid D_t \leq 0) \leq e^{-N\Delta^2/(16\sigma^2)} \leq \frac{\delta}{B}, \quad \mathbb{P}(\text{reject} \mid D_t \geq \Delta) \leq \frac{\delta}{B}. \quad (*)$$

Both directions come from the same Hoeffding bound: the measured gain deviates from D_t by $\geq \Delta/2$ with probability at most δ/B .

(i) *Soundness.* A false accept at step t is an acceptance with $D_t \leq 0$. By $(*)$ and a union bound over at most B decisions, the probability of any false accept during the run is at most δ . Call this good event E_1 .

(ii) *Liveness.* Let T be the first decision at which a candidate in G_ε is accepted; we bound T on a high-probability event. First observe that before such an acceptance the incumbent is never in G_ε : $a_0 \notin G_\varepsilon$ by hypothesis, and any accepted candidate either lies in G_ε (which would be the event in question) or does not (leaving the incumbent outside G_ε). Therefore, at every decision $t \leq T$, if the candidate lies in G_ε then (Δ, ε) -separation gives $D_t \geq \Delta$, and by $(*)$ the gate accepts it with probability at least $1 - \delta/B$, using evaluation randomness independent of $\mathcal{F}_{t-1}$ and of the proposal draw. Combining with Assumption 3.1,

$$\mathbb{P}(\text{decision } t \text{ accepts some } a \in G_\varepsilon \mid \mathcal{F}_{t-1}, T > t-1) \geq p_\gamma(1 - \frac{\delta}{B}) =: \rho.$$

By the peel-the-last-factor induction of Theorem 3.2(iii), $\mathbb{P}(T > t) \leq (1 - \rho)^t \leq e^{-\rho t}$, whence $\mathbb{E}[T] \leq 1/\rho$ and $\mathbb{P}(T > \ln(1/\delta)/\rho) \leq \delta$. Since $\delta \leq \frac{1}{2}$ and $B \geq 1$ give $\delta/B \leq \frac{1}{2}$, we have $\rho \geq p_\gamma/2$, so $T \leq \lceil 2 \ln(1/\delta)/p_\gamma \rceil$ with probability $\geq 1 - \delta$; the hypothesis $B \geq 2 \ln(1/\delta)/p_\gamma$ ensures the run is long enough. Call this good event E_2 .

(iii) *Cost, and conclusion.* On $E_1 \cap E_2$, which has probability $\geq 1 - 2\delta$ by the union bound, both (i) and (ii) hold, and the number of evaluator calls up to the certifying acceptance is at most $2N$ per decision times T decisions:

$$2NT \leq 2 \left\lceil \frac{16\sigma^2}{\Delta^2} \ln \frac{B}{\delta} \right\rceil \cdot \left\lceil \frac{2 \ln(1/\delta)}{p_\gamma} \right\rceil = O\left(\frac{\sigma^2}{\Delta^2} \cdot \frac{\ln(B/\delta) \ln(1/\delta)}{\gamma q_0(G_\varepsilon)}\right). \quad \square$$

Remark E.1. The argument uses Assumption 3.1 in exactly the form stated: almost surely over *all* histories, which in a gated run include accept/reject outcomes; this is what lets the boost survive the conditioning introduced by the gate. If incumbents are certified once and their means treated as known thereafter, Proposition 5.2(iii) halves the constant (8 for 16) and the evaluator cost in (iii) drops to NT .

F PROOFS FOR SECTION 7 (GENERALIZATION)

Proof of Theorem 7.1. Per-instance scores lie in $[0, 1]$, so for fixed a , Hoeffding's inequality (Hoeffding, 1963) gives $\mathbb{P}(|F(a) - \hat{F}_n(a)| \geq t) \leq 2e^{-2nt^2}$. Set $t_a = \sqrt{(\ell(a) \ln 2 + \ln(2/\delta))/(2n)}$ and union over $\mathcal{A}$:

$$\mathbb{P}(\exists a : |F(a) - \hat{F}_n(a)| \geq t_a) \leq \sum_{a \in \mathcal{A}} 2e^{-2nt_a^2} = \sum_{a \in \mathcal{A}} 2 \cdot 2^{-\ell(a)} \cdot \frac{\delta}{2} \leq \delta,$$

using the Kraft inequality $\sum_a 2^{-\ell(a)} \leq 1$ for prefix-free codes. The guarantee is simultaneous over $\mathcal{A}$, so it covers any data-dependent selection $\hat{a}$, however adaptive. $\square$

Proof of Corollary 7.2. Let a_L^* attain $\max_{\ell(a) \leq L} F(a)$ and let $\varepsilon_L = \sqrt{(\ell(a_L^*) \ln 2 + \ln(2/\delta))/(2n)}$. On the event of Theorem 7.1, every a with $\ell(a) \leq L$ has $|F(a) - \hat{F}_n(a)| \leq \varepsilon_L$. Hence

$$F(\hat{a}_L) \geq \hat{F}_n(\hat{a}_L) - \varepsilon_L \geq \hat{F}_n(a_L^*) - \varepsilon_L \geq F(a_L^*) - 2\varepsilon_L,$$

and $\sup_a F - F(\hat{a}_L) \leq (\sup_a F - F(a_L^*)) + 2\varepsilon_L = A(L) + 2\varepsilon_L$. $\square$

G PROOFS FOR SECTION 8 (PROXY GAPS)

Proof of Proposition 8.1. (i) $F(a_T) - F(a_0) = (\tilde{F}(a_T) - \tilde{F}(a_0)) - (b(a_T) - b(a_0)) \geq g - 2\alpha$.

(ii) For the upper bound, condition on the F -values and note $\tilde{F}(\hat{a}) - F(\hat{a}) = b(\hat{a}) \leq \max_{i \leq m} b_i$; the expected maximum of m independent σ_b -subgaussian variables is at most $\sigma_b \sqrt{2 \ln m}$ (the MGF argument of Theorem 5.1(i)). When all $F(a_i)$ are equal and $b_i \sim \mathcal{N}(0, \sigma_b^2)$ i.i.d., $\hat{a} = \arg \max b_i$, and the Gaussian extreme-value lower bound from Appendix C gives $\mathbb{E}[b(\hat{a})] \geq (1 - o(1)) \sigma_b \sqrt{2 \ln m}$.

(iii) The construction stays inside the model. Take $\mathcal{A} = \{a_0, a_1, a_2, \dots\}$ with

$$F(a_j) = 2^{-j} \in (0, 1], \quad \tilde{F}(a_j) = 1 - 2^{-(j+1)} \in [\tfrac{1}{2}, 1),$$

so the bias is $b(a_j) = \tilde{F}(a_j) - F(a_j) = 1 - 3 \cdot 2^{-(j+1)}$, which grows along the path—the formal meaning of an evaluator that the loop edits as it goes (each step is a one-line “relax the check” edit, so the proposer places mass on a_{j+1} from a_j). The proxy ratchet accepts every step since $\tilde{F}$ is strictly increasing; along the run, measured score rises to 1 while true score falls from $F(a_0) = 1$ to 0, traversing the full range. For the final claim: after T steps the measured progress is $g_T = \tilde{F}(a_T) - \tilde{F}(a_0) = \frac{1}{2} - 2^{-(T+1)} \in (0, \frac{1}{2})$ while the true progress is $F(a_T) - F(a_0) = 2^{-T} - 1 \rightarrow -1$. Any transfer inequality F -progress $\geq \psi(\tilde{F}$ -progress) with $\psi(g) > -1$ for some $g \in (0, \frac{1}{2})$ attained infinitely often is violated at large T ; only the vacuous bound $\psi \equiv -1$ (the full range) survives. (Documented instances of the hypothesis: hallucinated tool logs (Zhang et al., 2025a), simulator exploits in evolved rewards (Ma et al., 2024).) $\square$

H EXPERIMENTAL DETAILS AND AUDIT TRAIL

All simulations are pure NumPy, seeded (`default_rng(0)`), and complete in under a minute on one CPU core; the script `code/simulations.py` regenerates all figures and prints every numeric claim below. The RNG call order of the ratchet simulation is kept identical across revisions so that the seed-0 numbers are stable under audit.

Ratchet (Fig. 2). $T = 2000$ null candidates, Gaussian noise $\sigma = 1$, $R = 400$ independent runs. Panel (a): median and 10–90% band of m_t , against the $\sigma\sqrt{2 \ln t}$ upper bound. Realized values at $t = 2000$: median 3.39σ , mean 3.44σ , against the bound’s 3.90σ ; the deficit matches the second-order extreme-value correction $\approx (\ln \ln t + \ln 4\pi)/(2\sqrt{2 \ln t}) \approx 0.5\sigma$. Panel (b): mean cumulative accepts (8.30 simulated at $t = 2000$) vs. $H_{2000} = 8.18$. The gated variant implements the one-sided gate of Corollary 5.3(ii)—appropriate here because the incumbent (null) mean is known to be 0—with $N = \lceil 8\sigma^2 \ln(2000/0.05) \rceil = \lceil 84.77 \rceil = 85$ evaluations per candidate and threshold $\Delta/2$. Across the $400 \times 2000 = 8 \times 10^5$ gated decisions it accepted 2 null edits: a per-run rate of 0.005, within the $\delta = 5\%$ per-run certificate, and consistent with the exact per-candidate rate $p_N = \mathbb{P}(\mathcal{N}(0, 1/N) \geq \frac{1}{2}) \approx 2.0 \times 10^{-6}$, i.e. a Poisson prediction of ≈ 1.61 accepts in total. *Audit note:* an earlier draft reported this count as “zero across all runs,” an artifact of reading a single seed’s flat plot—in a paper about one-evaluation illusions, a one-seed illusion. The inset of Figure 2b now shows the rate on a log scale instead of letting it render as exactly zero.

Feedback (Fig. 3). $L \in \{2, \dots, 14\}$, 60 runs per point; *all three arms execute their algorithm*. Scalar: uniform search without replacement over an explicit random permutation of the 2^L candidates (mean hitting position $(2^L + 1)/2$). First-error: the prefix-fixing algorithm of Theorem 4.2(ii) from the all-zeros string, counting the final confirming query, hence average $\approx L/2 + 1$ for a uniform target (worst case $L + 1$; identification per se uses $\leq L$). Distance probing: the base-plus-flips scheme of Theorem 4.2(iii), executed with a correctness assertion on the reconstructed target; deterministically $L + 1$ queries.

Deception (Fig. 4). $\varepsilon = 0.3$; landscape $f(i) = k - i$ on $[0, k]$, $f(i) = 2(i - k)$ on $[k, 2k]$ (unique optimum, Theorem 5.4); proposals to $\{i - 1, i + 1\}$ uniformly, out-of-range rejected. Exact $\mathbb{E}[T]$ sums the first-step recurrences h_i over the full chain $[0, 2k]$ including the ascent side ($p = 1/2$, $q = \varepsilon/2$) and

the valley floor ($p = q = 1/2$); the acceptance probabilities, hence the exact curve, are unchanged by the asymmetric slope. Verification printouts: exact-to-bound ratio $\mathbb{E}[T]/(2\varepsilon^{-k}) \rightarrow 4.08$ at $k = 25$; archive means 111/420/2551 at $k = 5/10/25$ against the $4k(2k + 1)$ bounds 220/840/5100. Simulation: tolerant greedy $k \leq 10, 12$ runs per point (larger k is computationally pointless given the exact curve); archive 60 runs per point, $k \leq 25$.

I A PRE-REGISTERED PROTOCOL FOR MEASURING THE TRACE BOOST γ

Proposition 3.5 reduces Assumption 3.1 to an estimation problem. The following protocol instantiates it at rebuttal scale; we state it concretely enough to be falsified.

Task family. $M = 120$ small Python functions, each with a single planted bug and a deterministic test suite (3–10 unit tests; e.g. curated from HumanEvalFix-style corpora or hand-written). The good set G for each task is the set of patches under which all tests pass; membership is decided by the verifier (run the tests), not by any model.

Conditions. For each task, sample $K = 50$ independent patch proposals at temperature 1 from one frozen open-weights model under two prompts that differ *only* in feedback: (a) *cold*: the buggy function and the name of the failing test; (b) *trace*: additionally the full execution trace (stack trace, error message, failing assertion with values). Identical sampling parameters; no system-prompt differences beyond the feedback block.

Estimates. Per condition, the pooled pass rate $\hat{q} = (\#\text{passing proposals})/(MK)$ with a Wilson 95% interval (Brown et al., 2001), and the per-task rates for heterogeneity. The boost estimate is $\hat{\gamma} = \hat{q}_{\text{trace}}/\hat{q}_{\text{cold}}$ with the interval ratio; Proposition 3.5 gives the a-priori sample-size certificate (e.g. $q_{\min} = 0.05, \eta = 0.5, \delta = 0.05$ requires $K_{\text{tot}} \approx 7.9 \times 10^3$ verified proposals per condition, within the $MK = 6 \times 10^3$ budget up to the stated tolerance; tasks with $\hat{q}_{\text{cold}} = 0$ are reported separately as cold-unsolvable).

Outcomes. The framework predicts $\hat{\gamma}$ substantially above 1, with task-level $\hat{\gamma}$ correlating with trace informativeness; Theorem 3.2(iii) then converts $\hat{\gamma}$ into a hitting-time prediction checkable against the same logs. The falsifying outcome is an interval for $\hat{\gamma}$ containing 1: Assumption 3.1 would then be doing no work on this family, and the paper’s adaptive story would need revision. Total cost is roughly 1.2×10^4 generations plus test executions—one GPU-day with a small open model.

Scaling up. The same design lifts to harness search for web agents, where regenerable environments with per-task verifiers supply the fresh, repeatable evaluator the theory requires (Anonymous, 2026; Zhou et al., 2024): conditions become harness variants with and without execution-history feedback, and G is task success under the benchmark verifier.

J THE OPEN PROBLEM: RATES FOR SELF-REFERENCE

Let an outer loop optimize an artifact that is itself an optimizer (STOP’s improver, PromptBreeder’s mutation prompts, the DGM’s self-modifications), and let α_t denote the evaluator gap $\sup |b|$ at round t while m_t denotes the optimization pressure (candidates considered per round). Proposition 8.1(ii) suggests the harvested bias grows like $\sigma_b \sqrt{2 \ln m_t}$ while (i) caps transferable progress at $g_t - 2\alpha_t$.

Open problem: give natural conditions on the rates (α_t, m_t, g_t) —evaluator re-estimation on fresh data vs. pressure growth—under which the self-referential ratchet converges to true improvement rather than Goodhart divergence, and matching impossibility results. Even the one-level version is open in the form stated: the bandit literature controls α_t by design, and the RSI literature (Yampolskiy, 2015; Chalmers, 2010; Schmidhuber, 2003) discusses the limit without rates. We conjecture a phase transition at $\alpha_t = \Theta(\sigma_b \sqrt{\ln m_t})$.

K EXTENDED SYSTEM MAPPING

Table 1 compressed; here we record the fuller correspondence used in §2–§8. Sources marked informal in Table 1 are used as illustrations only.

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- **GEPA** (Agrawal et al., 2026): $\mathcal{A}$ = module prompts of a compound system; y = full execution traces plus judge text; sel = per-instance Pareto frontier (Theorem 5.4(iii)-style diversity preservation); minibatch gating (Proposition 5.2 with small N); reported test-set degradation when validation is reused (Theorem 7.1 with growing ℓ , Proposition 8.1(ii) with m = candidates tried).
 - **SkillOpt** (Yang et al., 2026): $\mathcal{A}$ = one skill document; bounded add/delete/replace edits (controlling $\ell(a)$ growth, Corollary 7.2); strict held-out acceptance (fresh-data re-estimation, Proposition 8.1(i)).
 - **ACE** (Zhang et al., 2025b): $\mathcal{A}$ = evolving context/playbook; documented *context collapse* and *brevity bias* are the two ends of Corollary 7.2’s U.
 - **Meta-Harness** (Lee et al., 2026): $\mathcal{A}$ = executable harness; y = raw prior code, traces, scores (maximal-information channel, §4); long-horizon credit assignment over $\mathcal{M}$.
 - **autoresearch** (Karpathy, 2026) (informal): $\mathcal{A}$ = training script; σ large (5-minute proxy runs), single-evaluation greedy ratchet—an in-the-wild instance of Theorem 5.1; reported surviving discoveries are those re-verified at higher N (Corollary 5.3). The archival grounding for this failure mode is Henderson et al. (2018); Dodge et al. (2019); Agarwal et al. (2021).
 - **Heuristic Learning** (Weng, 2026) (informal): $\mathcal{A}$ = executable heuristics; y = logs/video replays; “compress history” = explicit $\ell(a)$ control; reported failure on sparse-feedback exploration games = the misleading-feedback regime of Remark 3.4.
 - **ADAS / DGM** (Hu et al., 2025; Zhang et al., 2025a): $\mathcal{A}$ = agent programs; archives/lineages (Theorem 5.4(iii)); the DGM paper’s hallucinated-log episode (Proposition 8.1(iii)).
 - **FunSearch / AlphaEvolve** (Romera-Paredes et al., 2024; Novikov et al., 2025): $\mathcal{A}$ = programs; verified scalar evaluator ($\alpha = 0$, so Proposition 8.1(i) is vacuous-strong and scalar feedback’s slowness is paid in volume); island archives.
 - **Voyager** (Wang et al., 2024): additive skill library = monotone $\mathcal{M}$ growth re-weighting q_0 for later tasks (§3, reuse).
 - **Web/CUA harness optimization** (Anonymous, 2026; Zhou et al., 2024): regenerable environments with per-task verifiers supply the fresh, low- α , repeatable evaluator that Corollary 5.3 and Proposition 8.1(i) require; the framework predicts harness search on such benchmarks is the most theory-compliant instance of the loop currently available, and Appendix I uses it as the scaled-up venue for the γ measurement.

L LLM USAGE DISCLOSURE

In accordance with the ICLR policy on large language model usage: LLMs were used as general-purpose assistants for prose drafting, LaTeX preparation, and as coding assistants for the simulation scripts. All definitions, theorem statements, proofs, and numerical claims were checked by the authors; all simulation outputs regenerate deterministically from the committed seeded code, and the audit trail in Appendix H documents one numerical prose error caught and corrected during revision. No LLM is an author, and the authors take full responsibility for all content.